

Machine Learning (ML) - Theory Overview

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make decisions or predictions without being explicitly programmed for every task. ML systems improve their performance as they are exposed to more data. The primary goal is to build models that can generalize from training data to unseen data.

Types of Machine Learning:

Supervised Learning uses labeled data where the correct output is known. Common tasks include classification and regression.

Unsupervised Learning works with unlabeled data to discover hidden patterns, clusters, or relationships.

Reinforcement Learning trains an agent to make decisions through rewards and penalties while interacting with an environment.

Key Concepts and Applications

Important concepts in Machine Learning include datasets, features, labels, training, testing, model evaluation, overfitting, and underfitting. A dataset is divided into training and testing sets. The model learns from the training data and is evaluated on testing data to measure its performance.

Applications of Machine Learning include recommendation systems, image recognition, speech processing, healthcare diagnostics, fraud detection, autonomous vehicles, and natural language processing. As data availability continues to grow, Machine Learning plays a critical role in solving complex real-world problems and improving decision-making across industries.